

# Souvik De

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## Summary

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Backend Software Engineer with 4+ years of experience building production AI infrastructure, distributed backend systems, and developer tooling. Experienced in designing RAG platforms, document ingestion pipelines, vector search systems, and LLM orchestration using Go, TypeScript, and Python. Former Postman engineer, AsyncAPI maintainer, and Google Summer of Code mentor with experience building open-source tools used by thousands of developers.

## Experience

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**Software Engineer II**, XANE.AI – Gurgaon July 2025 – Present

- Built a multi-stage OCR → chunking → embedding → retrieval pipeline powering Maruti Suzuki's AI assistant, enabling semantic search across thousands of vehicle manuals.
- Designed and implemented an OCR-to-RAG ingestion pipeline converting unstructured PDFs into structured embeddings indexed in a vector database, enabling persistent knowledge memory for AI agents.
- Designed and implemented a scalable RAG evaluation pipeline using Ragas, enabling systematic measurement of retrieval accuracy, answer relevance, and overall system performance.

**Lead Software Engineer**, CodeMate.AI – Noida December 2024 – May 2025

- Engineered an enhanced search algorithm for Swagger files, optimizing RAG system performance and search precision.
- Implemented CI/CD pipeline utilizing GitHub Actions to automate application release cycles, enhancing deployment efficiency and reliability.
- Optimized a local server utilized by our VS Code extension, achieving over 30% improvement in startup time and enhancing overall user experience.
- Implemented automated deployment from GitHub Actions to a virtual machine (VM) and configured branch protection rules, improving code quality, deployment stability, and increasing team productivity by eliminating manual deployment efforts.

**Software Engineer**, Postman – Bangalore Feb 2022 – June 2024

- Developed and maintained a cross-platform CLI integrating official AsyncAPI tooling, providing a unified workflow for developers.
- Created and maintained AsyncAPI Bundler (16,000+ downloads), enabling reliable resolution of complex json ref dependency graphs across specification files.
- Served as maintainer across 3+ open-source projects, reviewing contributions, guiding architectural decisions, and driving roadmap initiatives.
- Mentored contributors under Google Summer of Code (GSoC), guiding projects to production-ready completion.
- Co-led governance initiative to standardize parser tooling architecture across multiple languages, improving maintainability and reducing duplication.

**Backend Developer Intern**, Mage – Remote Nov 2021 – Feb 2022

- Built microservices in Go to automate cloud infrastructure provisioning using templated configuration systems.
- Developed a React-based low-code interface for generating and provisioning infrastructure on demand.
- Deployed containerized services using Docker and Kubernetes in cloud-native environments.

## Projects

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**AI-powered GitHub Action for automated PR review summaries** Kitsu

- Built a GitHub Action that analyzes git patches using LLMs to generate concise pull request summaries.

- Handles large patch files by filtering low-signal changes and chunking oversized diffs.
- Designed to run entirely inside GitHub Actions with minimal latency.

#### **GitHub Action to sync GitHub Issues to Notion Database**

Notion-Board

- Developed a GitHub Action to auto sync and update GitHub issues and its state with Notion Pages.
- Earned recognition from the open-source community through project stars and valuable user feedback.
- Tools Used: TypeScript, GitHub Actions.

#### **FastAPI application that scrapes AsyncAPI documentation to create a RAG system and provides a chat API to answer user queries.**

AsyncAPI-RAG

- Built data scraper to scrape AsyncAPI spec and tools documentation from GitHub.
- Used LangChain to chunk Markdown data, generated embeddings with the Nomic embed model from Ollama, stored embeddings in a Qdrant database, and exposed a REST API to handle user queries and return contextual answers.
- Tools Used: Python, qdrant, Langchain, ollama, docker

## **Technologies**

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**Languages:** JavaScript, TypeScript, Go, Python, SQL

**Backend:** Node.js, FastAPI, Gin

**AI Systems:** RAG Architectures, Vector Database(Qdrant), LLM Integration, Langchain, Ollama

**Databases:** MySQL, Firebase

**DevOps:** Docker, GitHub Actions, CI/CD

## **Education**

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**Chandigarh University**, BE in Computer Science

June 2018 – June 2022

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, System Design